

Right-Triangle Trigonometry

ANSWER KEY



Trigonometric ratios

- 1 A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:

a $\sin \theta = \frac{5}{13}$

b $\cos \theta = \frac{12}{13}$

c $\tan \theta = \frac{5}{12}$

- 2 Write the exact value of each expression:

a $\sin 30^\circ = \frac{1}{2}$

b $\cos 45^\circ = \frac{\sqrt{2}}{2}$

c $\tan 60^\circ = \sqrt{3}$

d $\sin 60^\circ = \frac{\sqrt{3}}{2}$

- 3 A right triangle has legs of length 7 and 24.

a Find the length of the hypotenuse. $\sqrt{7^2 + 24^2} = \sqrt{625} = 25$.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree.
 $\theta = \tan^{-1}\left(\frac{7}{24}\right) \approx 16.26^\circ$.

Applications

- 4 A ladder leans against a wall, making an angle of 62° with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.
 $h = 4.5 \tan 62^\circ \approx 4.5 \times 1.8807 \approx 8.46$ m.

- 5 From the top of a cliff 50 m above sea level, the angle of depression to a boat is 18° . How far is the boat from the base of the cliff? Round to one decimal place.
 $d = \frac{50}{\tan 18^\circ} \approx \frac{50}{0.3249} \approx 153.9$ m.

- 6 A kite string is 85 m long and makes an angle of 54° with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.
 $h = 85 \sin 54^\circ \approx 85 \times 0.8090 \approx 69$ m.